

CAN 304

Computer Systems Security

Tutorial 8. Security Incidents Analysis

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Security Incident

A cyberattack disrupts vehicles fitted with anti-drunk driving systems, leaving hundreds unable to operate

Intoxalock, a US-based company, manufactures ignition interlock devices, breathalyzer systems installed in vehicles to prevent drunk driving. These devices are typically mandated for DUI offenders. To remain functional and legally compliant, [the devices require a mandatory calibration at authorized service centers every 25 to 30 days](#). If this maintenance is missed, the system automatically locks the vehicle as a safety precaution.

On March 14, 2026, Intoxalock suffered a Distributed Denial-of-Service (DDoS) attack. Attackers flooded the company's servers with massive amounts of traffic, causing a complete system shutdown. Because the system was down, [hundreds of drivers who were due for their routine calibration could not complete it, resulting in their vehicles being remotely locked](#). For example, in Connecticut alone, 7% to 10% of the company's 6,000 active users were directly impacted and lost mobility.

The company paused some systems as a precaution, then granted 10-day extensions, set up dedicated support channels, and covered direct costs. Services were restored on March 19.

Question 1

The interlock device requires periodic calibration. How did the attack exploit this dependency to amplify the impact?

- The attack disrupted the centralized calibration management system, preventing users from completing mandatory 25 to 30 day calibrations.
- Because the device locks automatically when calibration is overdue, the attack turned a routine dependency into a mass mobility failure. Hundreds of users in one state alone were directly stranded.

Question 2

The Intoxalock devices themselves are not directly internet-facing, but their calibration process depends entirely on the company's backend servers. In a DDoS attack, those servers became unreachable. For safety-critical IoT systems that require periodic online validation, how would you architect resilience against DDoS attacks?

- Architectural resilience against DDoS:
 - Redundant infrastructure: Multi-cloud deployment with auto-scaling and traffic scrubbing services (e.g., Cloudflare, AWS Shield).
 - Geographic distribution: Server clusters across multiple regions to prevent single-point failure.
 - Rate limiting and traffic filtering: Implement at network edge to drop malicious traffic before it reaches application servers.

Question 3

What possible motivations could attackers have for launching a DDoS attack against an ignition interlock company, given no data breach or ransom was publicly reported? How does this differ from ransomware motives?

- Possible motivations:
 - Protest against mandatory alcohol interlock laws or the company's business practices.
 - Disrupt Intoxalock to drive customers to competing interlock providers.
 - Ransom demand may have been made but not disclose
- Difference from ransomware:

	DDoS	Ransomware
Primary goal	Disrupt availability	Extort money via encryption
Data impact	data remained secure	Data encrypted or stolen
Recovery	Service restoration	Decryption key required
Visibility	Immediately noticeable	May be latent